Valori medii conditionate

Fie (X,Y) vect. aleat. discret

$$E(X=x_i/Y=y_j)=\sum_{i=1}^{u} x_i P(X=x_i/Y=y_j)$$

$$E(Y=y_j/X=x_i)=\sum_{j=1}^{m} y_j P(Y=y_j/X=x_i)$$

Fie (X,Y) vect. aleat. continuu cu densit. de probab. $f_{(X,Y)}(x,y)$

$$E(X=x/Y=y)=\int_{-\infty}^{\infty} x f(x/y)\,dx$$

$$E(Y=y/X=x)=\int_{-\infty}^{\infty} y f(y/x)\,dy$$

Momente

(1) Momentul initial de ordinul k

$$m_k=E(X^k)$$

Dacă X var. aleat. discretă $X\sim\begin{pmatrix} x_1 & \dots & x_u \\ p_1 & \dots & p_u \end{pmatrix}$

atunci $$m_k=E(X^k)=\sum_{i=1}^{u} x_i^k p_i$$

Dacă X var. aleat. continuă cu dens. de probab. $f_X(x)$ atunci

$$m_k = E(X^k) = \int_{-\infty}^{\infty} x^k \cdot f_X(x)\,dx$$

(2) Momentul absolut de ordinul k

$$\mu'_k = E(|X|^k)$$

(3) Momentul centrat de ordinul k

$$\mu_k = E((X - E(X))^k)$$

$$\mu_1 = E(X - E(X)) = E(X) - E(X) = 0$$

$$\mu_2 = E((X - E(X))^2) = D^2(X)$$

(4) Momentul centrat absolut de ordinul k

$$\mu'_k = E(|X - E(X)|^k)$$

Covariația și coeficientul de corelație

$$\text{cov}(X, Y) = E((X - E(X))(Y - E(Y)))$$

covariația var. aleat. X și Y

$\text{cov}(X,X) = E((X-E(X))^2) = D^2(X)$

$$\boxed{\text{cov}(aX,bY) = ab\,\text{cov}(X,Y)} \quad a,b \in \mathbb{R}^*$$

Demo $\text{cov}(aX,bY) = E((aX-E(aX))(bY-E(bY)))$

$= E((aX-aE(X))(bY-bE(Y)))$

$= E(ab(X-E(X))(Y-E(Y))) =$

$= ab\,E((X-E(X))(Y-E(Y))) = ab\,\text{cov}(X,Y)$

Th 1 $\boxed{\text{cov}(X,Y) = E(XY) - E(X)E(Y)}$

Demo $\text{cov}(X,Y) = E((X-E(X))(Y-E(Y)))$

$= E(XY - XE(Y) - YE(X) + E(X)E(Y))$

$= E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y)$

Th 2 Dacă X și Y sunt var. aleat. indep. atunci $\text{cov}(X,Y) = 0$

Demo X,Y independente $\xrightarrow[\text{med.}]{\text{prop.}} E(XY) = E(X)\cdot E(Y) \xrightarrow{\text{th1}}$

$\xrightarrow{\text{th1}} \text{cov}(X,Y) = 0$

$$\rho(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sqrt{D^2(X)\cdot D^2(Y)}}$$

coeficientul de corelatie al var. X și Y

Def Dacă $\rho(X,Y)=0$ sau $\operatorname{cov}(X,Y)=0$ atunci X și Y se numesc necorelate

Obs

Din tha $\Rightarrow$ dacă X,Y var. aleat. independente atunci X,Y sunt necorelate

!! Reciproca nu e adevărată !!

ex Fie $X \sim U[-1;1]$; $Y = X^2$. Sunt X și Y var. aleat. necorelate?

$$f_X(x) = \begin{cases} \frac{1}{2} & ; x \in [-1;1] \\ 0 & ; \text{în rest} \end{cases}$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_{-1}^{1} x \cdot \frac{1}{2}\,dx = 0$$

$$E(XY) = E(X^3) = \int_{-\infty}^{\infty} x^3 f_X(x)\,dx = \int_{-1}^{1} x^3 \cdot \frac{1}{2}\,dx = 0$$

$$\operatorname{cov}(X,Y) = \underbrace{E(X,Y)}_{=0} - \underbrace{E(X)}_{=0}\cdot E(Y) = 0 \Rightarrow X,Y \text{ var. aleat. necor.}$$

Th 3 Fie X, Y var. aleat. cu dispersii finite atunci $D^2(X+Y) = D^2(X) + D^2(Y) + 2\operatorname{cov}(X,Y)$

Demo $D^2(X+Y) = E((X+Y)^2) - (E(X+Y))^2$

$= E(X^2 + 2XY + Y^2) - (E(X) + E(Y))^2$

$= E(X^2) + 2E(XY) + E(Y^2) - (E(X))^2 - 2E(X)E(Y) - (E(Y))^2$

$= D^2(X) + D^2(Y) + 2\operatorname{cov}(X,Y)$

Generalizare Fie $(X_i)_{i=\overline{1,n}}$ var. aleat. cu dispersii finite atunci:

$$D^2\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} D^2(X_i) + 2 \sum_{\substack{i,j=1 \\ i \neq j}}^{n} \operatorname{cov}(X_i, X_j)$$

Th 4 $-1 \leq \rho(X,Y) \leq 1$

Demo Fie $W = X - aY \quad a \in \mathbb{R}^*$

$0 \leq D^2(W) = D^2(X - aY) = E((X-aY)^2) - (E(X-aY))^2$

$= E(X^2 - 2aXY + a^2Y^2) - (E(X) - aE(Y))^2$

$= E(X^2) - 2aE(XY) + a^2E(Y^2) - (E(X))^2 + 2aE(X)E(Y) + a^2(E(Y))^2$

$$= D^2(X) + a^2 D^2(Y) - 2a\,cov(X,Y) \Rightarrow$$

$$\Rightarrow 2a\,cov(X,Y) \leq D^2(X) + a^2 D^2(Y)$$

Notăm $\sigma_X = \sqrt{D^2(X)}$

$\sigma_Y = \sqrt{D^2(Y)}$

Luăm $a = \frac{\sigma_X}{\sigma_Y}$

$$2 \cdot \frac{\sigma_X}{\sigma_Y} cov(X,Y) \leq \sigma_X^2 + \frac{\sigma_X^2}{\sigma_Y^2}\sigma_Y^2 \Rightarrow 2 \cdot \frac{\sigma_X}{\sigma_Y} cov(X,Y) \leq 2\sigma_X^2$$

$$\Rightarrow \frac{cov(X,Y)}{\sigma_X \sigma_Y} \leq 1 \Rightarrow \rho(X,Y) \leq 1$$

Luăm $a = -\frac{\sigma_X}{\sigma_Y}$

$$-2 \cdot \frac{\sigma_X}{\sigma_Y} cov(X,Y) \leq \sigma_X^2 + \frac{\sigma_X^2}{\sigma_Y^2}\sigma_Y^2 \Rightarrow -\frac{cov(X,Y)}{\sigma_X \sigma_Y} \leq 1$$

$$\Rightarrow \rho(X,Y) \geq -1$$

ex Să se scrie repartiția vectorului (completați tabelul). Sunt var. aleat X, Y independente? Dar necorelate?

$X \backslash Y$	-1	0	1	$P(X=x_i)$
-1	0	$0,25$	0	$0,25$
1	$0,25$	$0,25$	$0,25$	$0,75$
$P(Y=y_j)$	$0,25$	$0,5$	$0,25$	1

(y_j — coloana; x_i — linia; p_i — $P(X=x_i)$; q_j — $P(Y=y_j)$)

$p_{ij} = p_i q_j \quad \forall i,j \Rightarrow X, Y$ independ

$0 = p_{11} \neq p_1 \cdot q_1 = 0{,}25 \cdot 0{,}25 \Rightarrow X, Y$ sunt independ.

$$X \sim \begin{pmatrix} -1 & 1 \\ 0{,}25 & 0{,}75 \end{pmatrix} \qquad Y \sim \begin{pmatrix} -1 & 0 & 1 \\ 0{,}25 & 0{,}5 & 0{,}25 \end{pmatrix}$$

$$XY \sim \begin{pmatrix} -1 & 0 & 1 \\ 0{,}25 & 0{,}5 & 0{,}25 \end{pmatrix}$$

$E(X) = -1 \cdot 0{,}25 + 1 \cdot 0{,}75 = 0{,}5$

$E(Y) = -1 \cdot 0{,}25 + 0 \cdot 0{,}5 + 1 \cdot 0{,}25 = 0$

$E(XY) = -1 \cdot 0{,}25 + 0 \cdot 0{,}5 + 1 \cdot 0{,}25 = 0$

$cov(X,Y) = E(XY) - E(X) \cdot E(Y) = 0 \Rightarrow X, Y$ sunt necorelate

ex

$X \backslash Y$	-1	1	$P(X = x_i)$
-2	$0{,}4$	$0{,}1$	$0{,}5$
2	$0{,}1$	$0{,}4$	$0{,}5$
$P(Y = y_j)$	$0{,}5$	$0{,}5$	1 / 1

Aflati repart. vector. (X,Y), $D^2(X)$, $D^2(Y)$, $\rho(X,Y)$, $\rho(X+Y)$

$$X \sim \begin{pmatrix} -2 & 2 \\ 0{,}5 & 0{,}5 \end{pmatrix} \qquad Y \sim \begin{pmatrix} -1 & 1 \\ 0{,}5 & 0{,}5 \end{pmatrix}$$

$D^2(X) = E(X^2) - (E(X))^2$

$E(Y) = -1 \cdot 0{,}5 + 1 \cdot 0{,}5 = 0$

$E(X) = -2 \cdot 0,5 + 2 \cdot 0,5 = 0$ $\quad E(Y^2) = (-1)^2 \cdot 0,5 + 1^2 \cdot 0,5 = 1$

$E(X^2) = (-2)^2 \cdot 0,5 + 2^2 \cdot 0,5 = 4$ $\quad D^2(Y) = 1$

$D^2(X) = 4 - 0 = 4$

$$\operatorname{cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$XY \sim \begin{pmatrix} -2 & 2 \\ 0,2 & 0,8 \end{pmatrix} \Rightarrow E(X,Y) = -2 \cdot 0,2 + 2 \cdot 0,8 = 1,2$$

$$\operatorname{cov}(X,Y) = 1,2$$

$$\rho(X,Y) = \frac{\operatorname{cov}(X,Y)}{\sqrt{D^2(X) D^2(Y)}} = \frac{1,2}{2} = 0,6$$

$$D^2(X+Y) \overset{th\,3}{=\!=} D^2(X) + D^2(Y) + 2\operatorname{cov}(X,Y) = 4 + 1 + 2 \cdot 1,2 = 7,4$$

ex Fie X, Y 2 var. aleat. cu $E(X) = -2$; $E(Y) = 3$; $D^2(X) = 4$; $D^2(Y) = 5$; $\rho(X,Y) = -\frac{1}{2}$. Să se determine $E(Z) = ?$ unde $Z = 2X - XY + Y^2 - 2$.

$$E(Z) = E(2X - XY + Y^2 - 2)$$

$$= 2E(X) - E(XY) + E(Y^2) - E(2)$$

$$D^2(Y) = E(Y^2) - (E(Y))^2 \Rightarrow E(Y^2) = 5 + 9 = 14$$

$$\rho(X,Y)=\frac{E(XY)-E(X)E(Y)}{\sqrt{D^2(X)\cdot D^2(Y)}} \Rightarrow -\frac{1}{2}=\frac{E(XY)-(-2)\cdot 3}{\sqrt{4\cdot 5}}$$

$$\Rightarrow E(XY)=-\frac{2\cdot\sqrt{5}}{2}-6=-\sqrt{5}-6$$

$$E(Z)=2\cdot(-2)-(-\sqrt{5}-6)+14-2=-4+\sqrt{5}+6+14-2$$

$$E(Z)=14+\sqrt{5}$$

ex (X,Y) vect aleat. cu dens. de probab.

$$f_{(X,Y)}(x,y)=\begin{cases} kxy\,; (x,y)\in\{(x,y)\mid x\geq 0; y\geq 0; x+y\leq 1\}=\Delta \\ 0\,; \text{în rest}\end{cases}$$

a) $k=?$ a.î. f dens. de probab a vect. (X,Y)

$$\iint_D f_{(X,Y)}(x,y)\,dx\,dy=1 \Rightarrow \int_0^1\int_0^{1-x} kxy\,dy\,dx=1$$

$$\Rightarrow k\int_0^1 x\cdot\frac{y^2}{2}\Big|_0^{1-x}dx=1 \Rightarrow \frac{k}{2}\int_0^1 x(1-x)^2dx=1$$

$$\Rightarrow \frac{k}{2}\int_0^1 x-2x^2+x^3dx=1 \Rightarrow \frac{k}{2}\left(\frac{x^2}{2}-2\cdot\frac{x^3}{3}+\frac{x^4}{4}\right)\Big|_0^1=1$$

$$\Rightarrow \frac{k}{2}\left(\frac{1}{2}-\frac{2}{3}+\frac{1}{4}\right)=1 \Rightarrow \frac{k}{2}\cdot\frac{6-8+3}{12}=1 \Rightarrow k=24$$

b) $E(x), E(y), D^2(x), D^2(y), \rho(x,y)$

$$E(x) = \int_{-\infty}^{\infty} x f_x(x)\,dx$$

$$f_x(x) = \int_{-\infty}^{\infty} f_{(x,y)}(x,y)\,dy = \begin{cases} \int_0^{1-x} 24xy\,dy & ; x \in [0,1] \\ 0 & ; \text{ce rest} \end{cases}$$

$$f_x(x) = \begin{cases} 24 \cdot x \cdot \frac{y^2}{2}\Big|_0^{1-x} & ; x \in [0,1] \\ 0 & ; \text{în rest} \end{cases} = \begin{cases} 12x(1-x)^2 & ; x \in [0,1] \\ 0 & ; \text{în rest} \end{cases}$$

$$E(x) = \int_0^1 x \cdot 12x(1-x)^2\,dx = 12\int_0^1 x^2(1-2x+x^2)\,dx$$

$$= 12\left(\frac{x^3}{3} - 2\cdot\frac{x^4}{4} + \frac{x^5}{5}\right)\Big|_0^1 = 12\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) = \frac{2}{5}$$

$$E(y) = \int_{-\infty}^{\infty} y \cdot f_y(y)\,dy$$

$$f_y(y) = \int_{-\infty}^{\infty} f_{(x,y)}(x,y)\,dx = \begin{cases} \int_0^{1-y} 24xy\,dx & y \in [0,1] \\ 0 ; \text{ în rest} \end{cases}$$

$$= \begin{cases} \cancel{24}^{12} y \cdot \frac{x^2}{\cancel{2}}\Big|_0^{1-y} ; y \in [0,1] \\ 0 ; \text{ în rest} \end{cases} = \begin{cases} 12y\cdot(1-y)^2 ; y \in [0,1] \\ 0 ; \text{ în rest} \end{cases}$$

$$E(y)=\int_0^1 y\cdot 12y(1-y)^2\,dy=12\int_0^1 y^2-2y^3+y^4\,dy$$

$$=12\left(\frac{y^3}{3}-2\cdot\frac{y^4}{4}+\frac{y^5}{5}\right)\Big|_0^1=12\left(\frac{1}{3}-\frac{1}{2}+\frac{1}{5}\right)=\frac{2}{5}$$

$$E(x^2)=\int_{-\infty}^{\infty}x^2 f_x(x)\,dx=\int_0^1 x^2\cdot 12x(1-x)^2\,dx$$

$$=12\left(\frac{x^4}{4}-2\cdot\frac{x^5}{5}+\frac{x^6}{6}\right)\Big|_0^1=12\cdot\frac{15-24+10}{60}$$

$$=\frac{12}{60}=\frac{1}{5}$$

$$D^2(x)=E(x^2)-(E(x))^2=\frac{1}{5}-\left(\frac{2}{5}\right)^2=\frac{1}{5}-\frac{4}{25}=\frac{1}{25}$$

$$E(y^2)=\int_{-\infty}^{\infty}y^2 f_y(y)\,dy=\int_0^1 y^2\cdot 12y(1-y)^2\,dy=\frac{1}{5}$$

$$D^2(y)=\frac{1}{25}$$

$$E(xy)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}xy\,f_{(x,y)}(x,y)\,dx\,dy$$

$$=\iint_D xy\cdot 24xy\,dx\,dy=24\int_0^1\int_0^{1-x}x^2y^2\,dy\,dx$$

$$= 24 \int_0^1 x^2 \cdot \frac{y^3}{3}\Big|_0^{1-x} dx = 8 \int_0^1 x^2 (1-x)^3 dx =$$

$$= 8 \int_0^1 x^2 (1 - 3x + 3x^2 - x^3) dx = 8 \left(\frac{x^3}{3} - 3 \cdot \frac{x^4}{4} + 3 \cdot \frac{x^5}{5} - \frac{x^6}{6} \right) \Big|_0^1$$

$$= 8 \left(\frac{1}{3} - \frac{3}{4} + \frac{3}{5} - \frac{1}{6} \right) = 8 \frac{20 - 45 + 36 - 10}{60} = \frac{8}{60} = \frac{2}{15}$$

$$\rho(X,Y) = \frac{E(XY) - E(X)E(Y)}{\sqrt{D^2(X) D^2(Y)}} = \frac{\frac{2}{15} - \frac{4}{25}}{\sqrt{\frac{1}{25} \cdot \frac{1}{25}}} =$$

$$= \frac{\frac{10 - 12}{75}}{\frac{1}{25}} = \frac{-2}{3}$$

ex Se dă un vector aleator cu densitatea de probab. $f_{(X,Y)}(x,y) = \begin{cases} e^{-y}; & 0 \le x \le y \\ 0; & \text{în rest} \end{cases}$.

Calculați $E(X/Y=y)$; $E(Y/X=x)$

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\, dy = \begin{cases} \int_x^{\infty} e^{-y} dy & x \ge 0 \\ 0, \text{ în rest} \end{cases}$$

$$= \begin{cases} -e^{-y}\Big|_x^{\infty} & x \ge 0 \\ 0 \text{ în rest} \end{cases} = \begin{cases} e^{-x}, & x \ge 0 \\ 0; & \text{în rest} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y)\,dx = \begin{cases} \int_0^y e^{-y}\,dx & y \geq 0 \\ 0; \text{ in rest} \end{cases}$$

$$= \begin{cases} ye^{-y}; y \geq 0 \\ 0; \text{ in rest} \end{cases}$$

$$f(x/y) = \frac{f_{(X,Y)}(x,y)}{f_Y(y)} = \begin{cases} \frac{1}{y}; 0 \leq x \leq y \\ 0; \text{ in rest} \end{cases}$$

$$f(y/x) = \frac{f_{(X,Y)}(x,y)}{f_X(x)} = \begin{cases} e^{-y+x}; 0 \leq x \leq y \\ 0; \text{ in rest} \end{cases}$$

$$E(X/Y=y) = \int_{-\infty}^{\infty} x f(x/y)\,dx = \int_0^y x \cdot \frac{1}{y}\,dx = \frac{1}{y} \cdot \frac{x^2}{2}\Big|_0^y = \frac{y}{2}$$

$$E(Y/X=x) = \int_{-\infty}^{\infty} y f(y/x)\,dy = \int_x^{\infty} y e^{-y+x}\,dy =$$

$$= e^x\left(-ye^{-y}\Big|_x^{\infty} + \int_x^{\infty} e^{-y}\,dy\right) = e^x\left(xe^{-x} + e^{-y}\Big|_x^{\infty}\right)$$

$$= e^x\left(xe^{-x} + e^{-x}\right) = x + 1$$

ex La o petrecere cu $n \geq 2$ persoane fiecare persoană aruncă o pălărie într-o cutie. Cutia se amestecă și fiecare persoană extrage cu ochii închiși câte o pălărie din cutie fără a o pune înapoi.

Vom spune că are loc o potrivire dacă o persoană extrage din cutie propria pălărie.

a) Aflați media și dispersia var. aleat. V_n ce reprezintă numărul de potriviri

$$V_n = \sum_{i=1}^{n} X_i \qquad X_i \sim \begin{pmatrix} 0 & 1 \\ 1-\frac{1}{n} & \frac{1}{n} \end{pmatrix}$$

$$E(X_i) = 0 \cdot \left(1-\frac{1}{n}\right) + 1 \cdot \frac{1}{n} = \frac{1}{n}$$

$$E(V_n) = E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E(X_i) = \sum_{i=1}^{n} \frac{1}{n} = 1$$

$$E(X_i^2) = 0^2 \cdot \left(1-\frac{1}{n}\right) + 1^2 \cdot \frac{1}{n} = \frac{1}{n}$$

$$D^2(X_i) = E(X_i^2) - (E(X_i))^2 = \frac{1}{n} - \frac{1}{n^2} = \frac{n-1}{n^2}$$

$$\text{cov}(X_i, X_j) = E(X_i, X_j) - E(X_i) \cdot E(X_j)$$

$$X_i \cdot X_j \sim \begin{pmatrix} 0 & 1 \\ 1-\frac{1}{n(n-1)} & \frac{1}{n(n-1)} \end{pmatrix}$$

$$E(X_i X_j) = 0 \cdot \left(1 - \frac{1}{n(n-1)}\right) + 1 \cdot \frac{1}{n(n-1)} = \frac{1}{n(n-1)}$$

$$\operatorname{cov}(X_i, X_j) = \frac{1}{n(n-1)} - \frac{1}{n} \cdot \frac{1}{n}$$

$$D^2(V_n) = D^2\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} D^2(X_i) + 2 \cdot \sum_{i \neq j}^{n} \operatorname{cov}(X_i, X_j) =$$

$$= \sum_{i=1}^{n} \frac{n-1}{n^2} + 2 \cdot \sum_{i \neq j}^{n} \left(\frac{1}{n(n-1)} - \frac{1}{n^2}\right) = \frac{n-1}{n} + 2\left(\frac{1}{n(n-1)} - \frac{1}{n^2}\right) \cdot C_n^2$$

$$D^2(V_n) = \frac{n-1}{n} + \cancel{2} \cdot \frac{n!}{\cancel{2!}(n-2)!}\left(\frac{1}{n(n-1)} - \frac{1}{n^2}\right)$$

$$= \frac{n-1}{n} + n(n-1)\left(\frac{1}{n(n-1)} - \frac{1}{n^2}\right) = \frac{n-1}{n} + 1 - \frac{n-1}{n} = 1$$

b) Presupunem că fiecare persoană după extragere pune pălăria înapoi. Calculați și în acest caz media și dispersia lui V_n

$$E(V_n) = 1$$

$$D^2(V_n) = \sum_{i=1}^{n} D^2(X_i) = \sum_{i=1}^{n} \frac{n-1}{n^2} = \frac{n-1}{n}$$